

COURIER MANAGEMENT SYSTEM (CMS)

Project Overview

The **Courier Management System (CMS)** is a Salesforce-based cloud application designed to automate courier and logistics operations. The system helps manage shipment booking, parcel tracking, delivery agent assignment, invoice generation, and reporting through a centralized platform. It replaces manual record-keeping with an integrated digital system that provides real-time updates, improves operational efficiency, and enhances customer satisfaction. By using Salesforce features such as custom objects, automation flows, reports, and dashboards, the system enables courier companies to manage deliveries more efficiently and monitor business performance across multiple branches.

Problem Statements

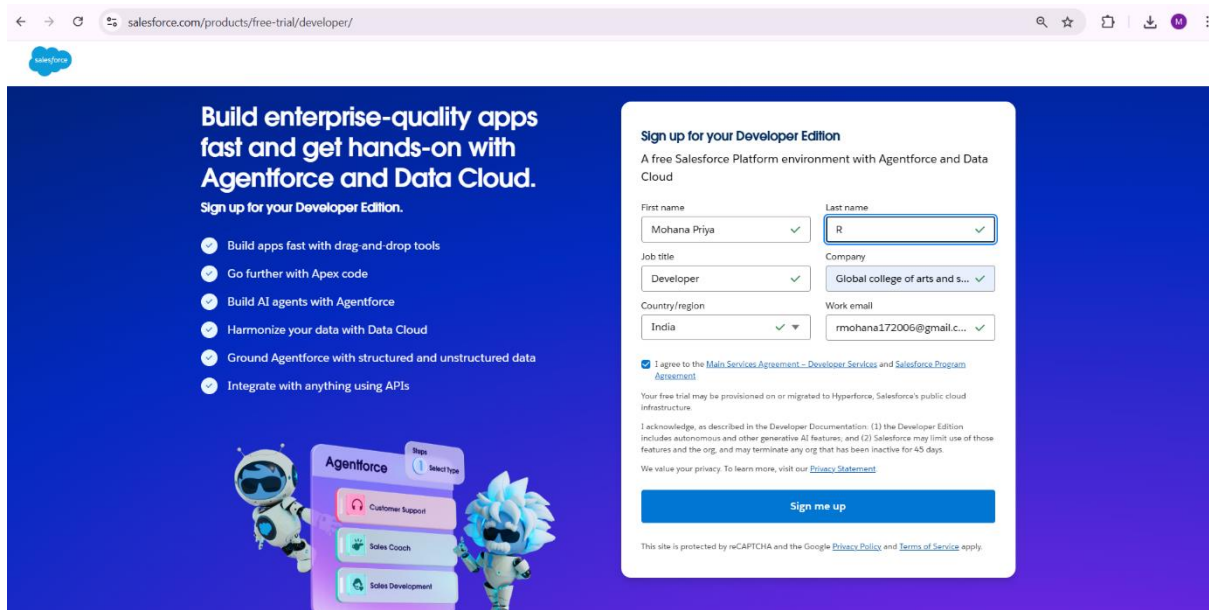
1. Courier companies rely on **manual registers and spreadsheets** to manage shipments, which leads to errors and delays.
2. Customers **cannot track parcel status in real time**, creating uncertainty about delivery progress.
3. Delivery agent assignments are done **manually**, causing scheduling conflicts and operational inefficiencies.
4. Lack of **centralized data management** makes it difficult for companies to manage multiple branches effectively.
5. Courier companies do not have **analytics or performance reports** to monitor delivery operations and revenue.

Planning Status (Project Planning)

The project was developed using the **Agile methodology**, where the work was divided into multiple sprints to ensure efficient development and testing. Each sprint focused on specific tasks such as system setup, data modeling, automation, security configuration, and report creation. The planning included creating user stories, assigning story points, prioritizing tasks, and tracking progress using sprint schedules. This approach helped the team manage development effectively, monitor progress, and deliver the Courier Management System in an organized and structured manner.

Milestone 1 – Salesforce Account

- A Salesforce Developer Account was created to develop the Courier Management System.
- The account was verified and activated using email confirmation.
- This account provides access to Salesforce tools such as objects, flows, reports, and dashboards.
- It acts as the main platform for building and managing the application.



The screenshot shows the Salesforce Developer Edition sign-up page. The page has a dark blue header with the Salesforce logo and navigation links. The main content area is white with a blue background. On the left, there's a section titled "Build enterprise-quality apps fast and get hands-on with Agentforce and Data Cloud." followed by "Sign up for your Developer Edition." and a list of benefits: "Build apps fast with drag-and-drop tools", "Go further with Apex code", "Build AI agents with Agentforce", "Harmonize your data with Data Cloud", "Ground Agentforce with structured and unstructured data", and "Integrate with anything using APIs". Below this is a "Steps" section with "1 Select type" and a "Customer Support" button. On the right, there's a "Sign up for your Developer Edition" form. The form includes fields for "First name" (Mohana Priya), "Last name" (R), "Job title" (Developer), "Company" (Global college of arts and s...), "Country/region" (India), and "Work email" (rmohana172006@gmail.c...). There's a checkbox for "I agree to the Main Services Agreement - Developer Services and Salesforce Program Agreement". Below the form, there's a "Sign me up" button and a note about the free trial and public cloud infrastructure. At the bottom, there's a disclaimer about the use of AI features and a link to the Privacy Statement.

salesforce.com/products/free-trial/developer/

Build enterprise-quality apps fast and get hands-on with Agentforce and Data Cloud.

Sign up for your Developer Edition.

- ✓ Build apps fast with drag-and-drop tools
- ✓ Go further with Apex code
- ✓ Build AI agents with Agentforce
- ✓ Harmonize your data with Data Cloud
- ✓ Ground Agentforce with structured and unstructured data
- ✓ Integrate with anything using APIs

Steps
1 Select type

Customer Support

Sales Coach

Sales Development

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data Cloud

First name: Mohana Priya ✓

Last name: R ✓

Job title: Developer ✓

Company: Global college of arts and s... ✓

Country/region: India ✓

Work email: rmohana172006@gmail.c... ✓

☒ I agree to the [Main Services Agreement - Developer Services and Salesforce Program Agreement](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

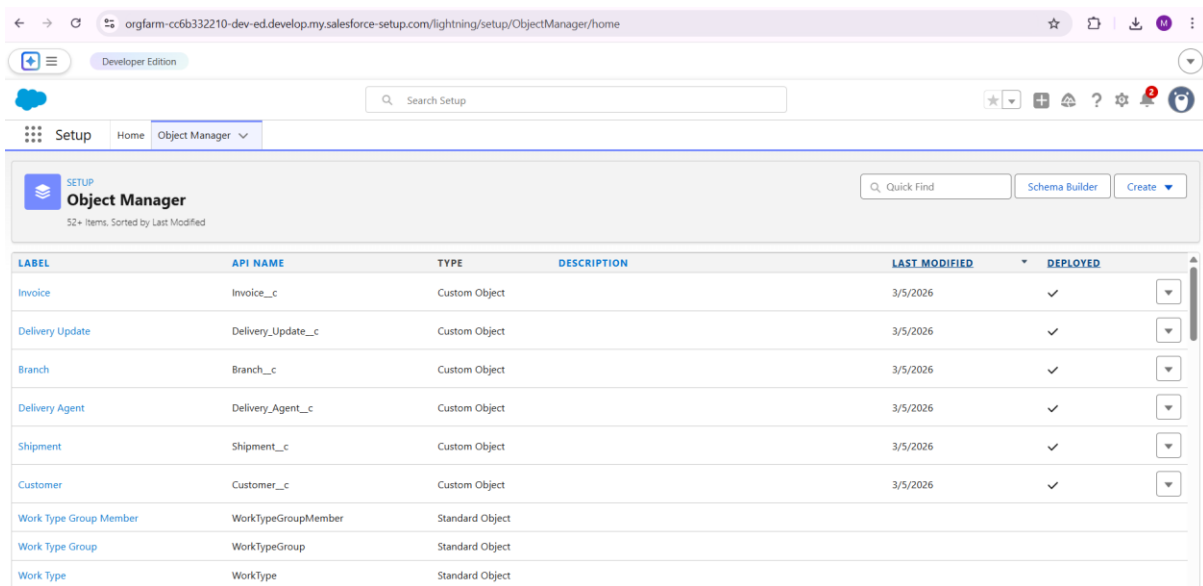
We value your privacy. To learn more, visit our [Privacy Statement](#)

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Milestone 2 – Object Creation

- Custom objects were created to store courier-related data.
- Objects include **Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice.**
- Each object works like a database table to store records.
- These objects help organize and manage information efficiently.

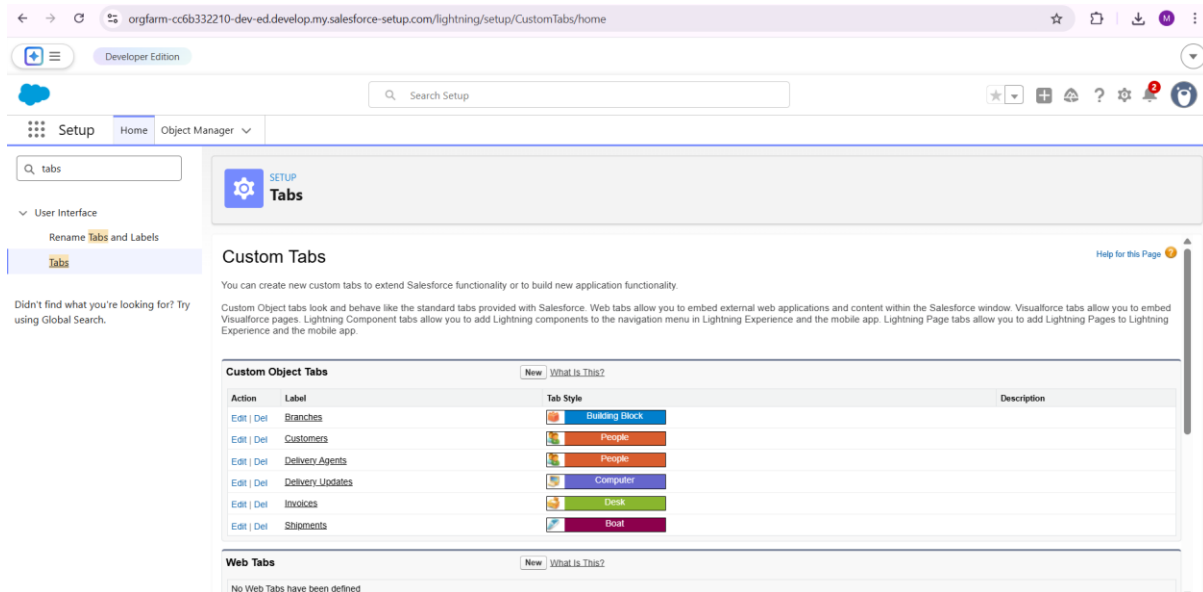


The screenshot shows the Salesforce Object Manager interface. The browser address bar displays the URL: orgfarm-cc6b332210-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/home. The page header includes the Salesforce logo, a search bar, and navigation links for Setup, Home, and Object Manager. The main content area is titled 'Object Manager' and shows a list of 52+ items, sorted by Last Modified. The table below lists the objects:

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Invoice	Invoice__c	Custom Object		3/5/2026	✓
Delivery Update	Delivery_Update__c	Custom Object		3/5/2026	✓
Branch	Branch__c	Custom Object		3/5/2026	✓
Delivery Agent	Delivery_Agent__c	Custom Object		3/5/2026	✓
Shipment	Shipment__c	Custom Object		3/5/2026	✓
Customer	Customer__c	Custom Object		3/5/2026	✓
Work Type Group Member	WorkTypeGroupMember	Standard Object			
Work Type Group	WorkTypeGroup	Standard Object			
Work Type	WorkType	Standard Object			

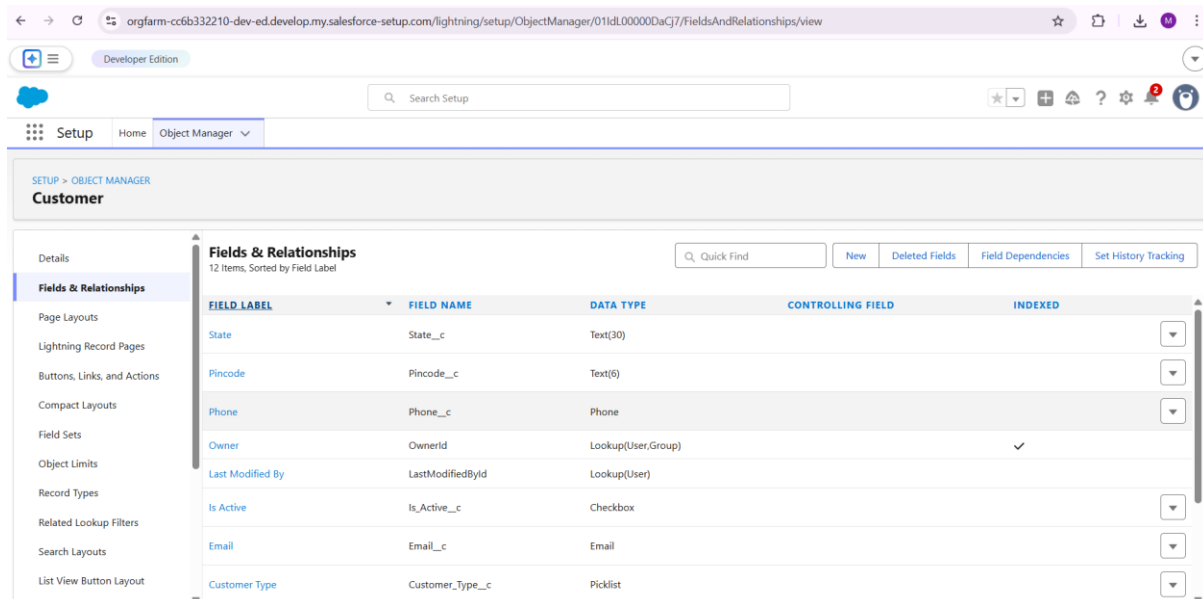
Milestone 3 – Tabs Creation

- Tabs were created for each custom object in Salesforce.
- Tabs allow users to easily access object data from the navigation menu.
- This improves usability and helps users quickly find information.
- Tabs were created for Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice.



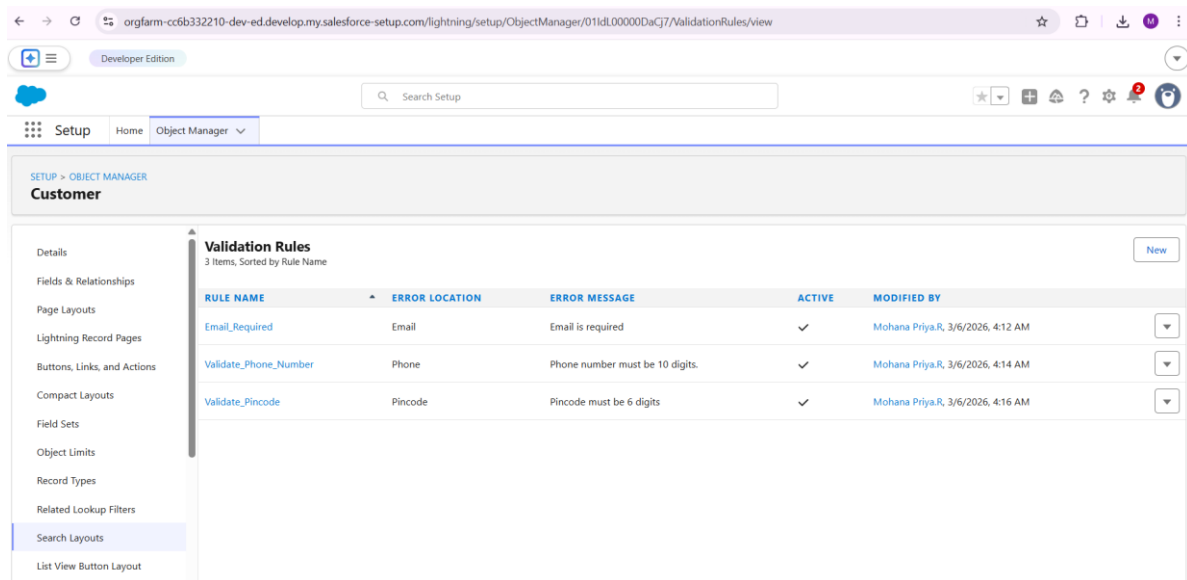
Milestone 4 – Fields and Relationships

- Fields were added to objects to store detailed information such as **email, phone number, address, shipment status, and delivery date**.
- Different field types such as text, number, date, picklist, and lookup were used.
- Relationships were created between objects to connect related data.
- For example, Shipment is linked with Customer and Delivery Agent.



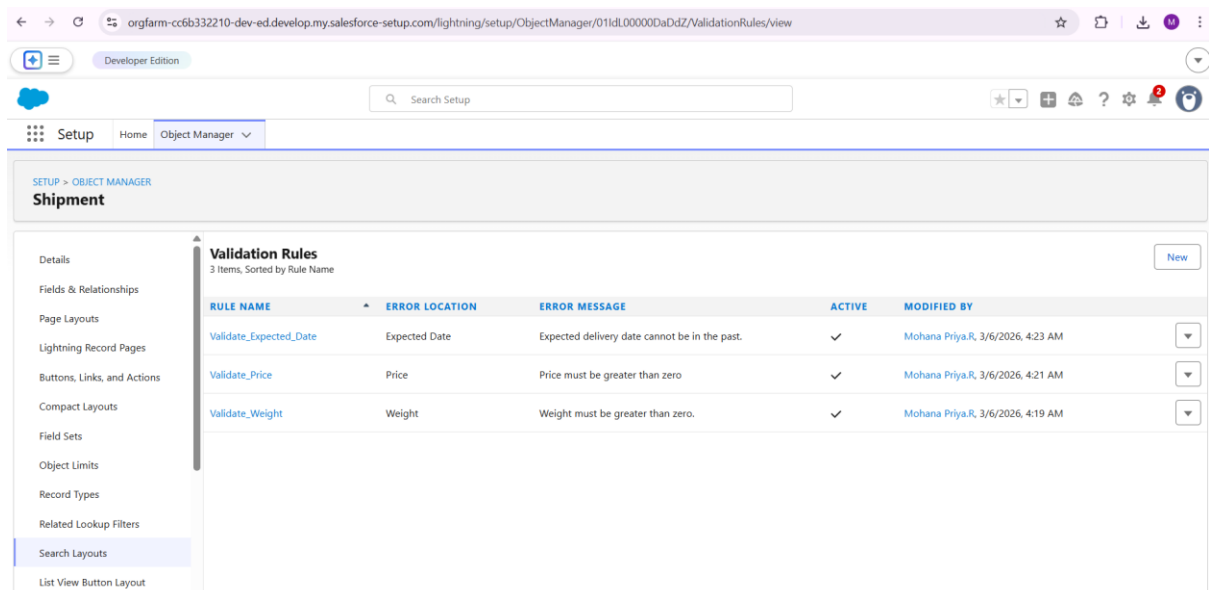
Milestone 5 – Validation Rules

- Validation rules were implemented to ensure accurate data entry.
- These rules prevent users from saving incorrect or incomplete records.
- Examples include checking phone number length and ensuring shipment weight is greater than zero.
- This improves data quality and system reliability.



The screenshot shows the Salesforce Setup interface for a Customer object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The main content area is titled 'Validation Rules' and shows 3 items, sorted by Rule Name. The table lists three validation rules:

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Email_Required	Email	Email is required	✓	Mohana Priya.R, 3/6/2026, 4:12 AM
Validate_Phone_Number	Phone	Phone number must be 10 digits.	✓	Mohana Priya.R, 3/6/2026, 4:14 AM
Validate_Pincode	Pincode	Pincode must be 6 digits	✓	Mohana Priya.R, 3/6/2026, 4:16 AM

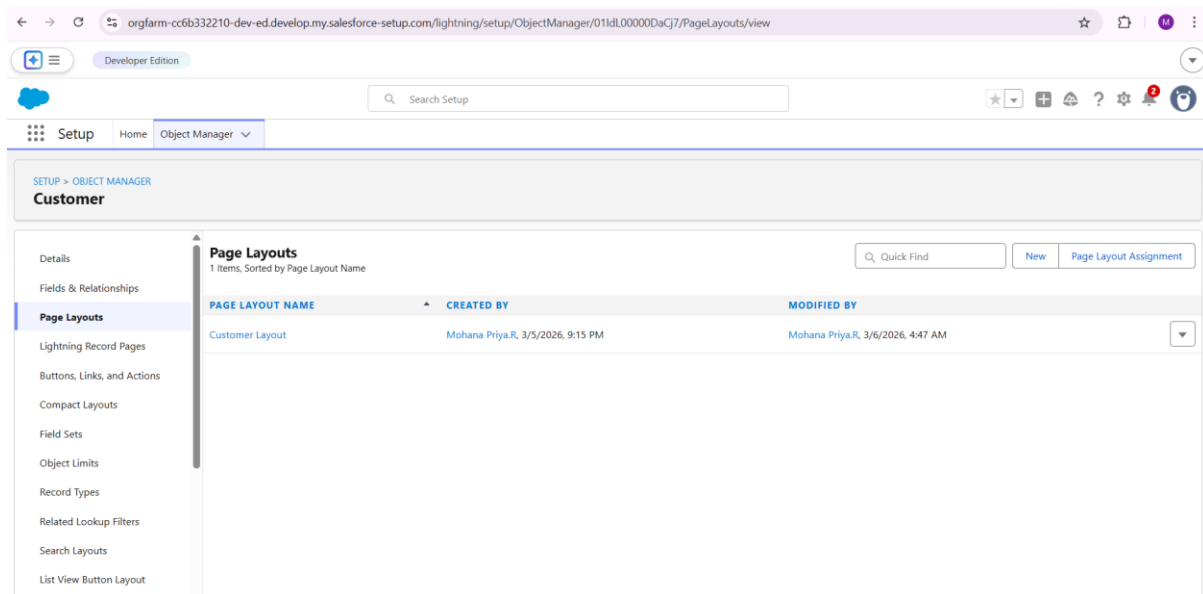


The screenshot shows the Salesforce Setup interface for a Shipment object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The main content area is titled 'Validation Rules' and shows 3 items, sorted by Rule Name. The table lists three validation rules:

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Validate_Expected_Date	Expected Date	Expected delivery date cannot be in the past.	✓	Mohana Priya.R, 3/6/2026, 4:23 AM
Validate_Price	Price	Price must be greater than zero	✓	Mohana Priya.R, 3/6/2026, 4:21 AM
Validate_Weight	Weight	Weight must be greater than zero.	✓	Mohana Priya.R, 3/6/2026, 4:19 AM

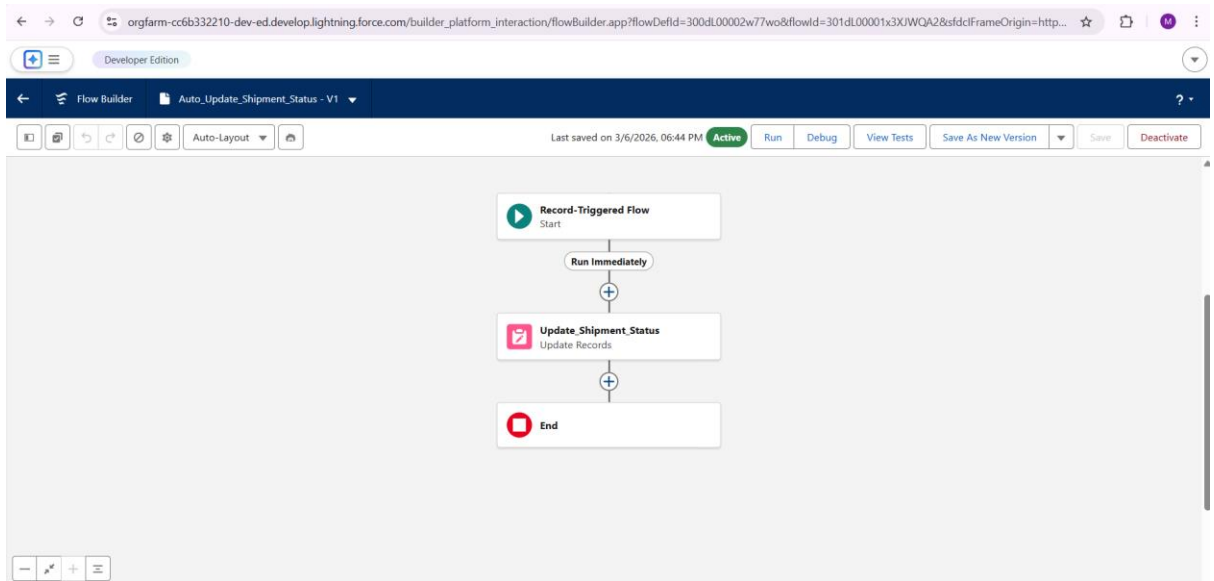
Milestone 6 – Page Layouts

- Page layouts were customized to organize fields and sections properly.
- Important fields were grouped into sections like customer details, shipment details, and address information.
- This improves the user interface and makes data entry easier.
- Users can quickly view and update necessary information.



Milestone 7 – Flows (Automation)

- Automation flows were created using Salesforce Flow Builder.
- The system automatically updates shipment status when a delivery update is created.
- This reduces manual work and improves efficiency.
- Automation ensures faster processing of courier operations.



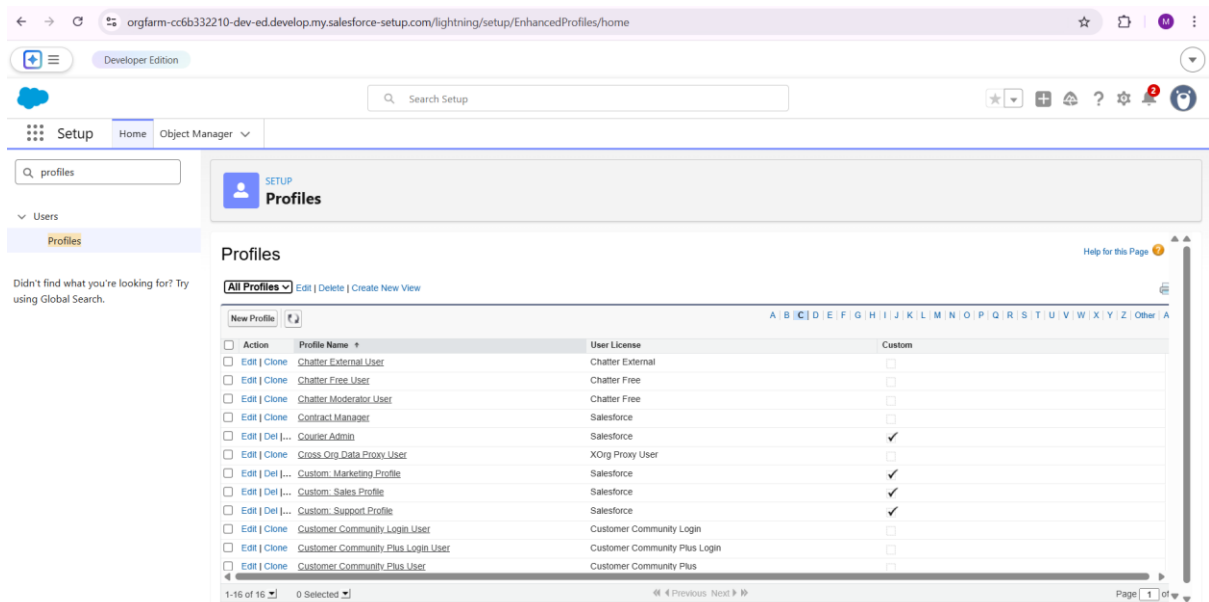
Milestone 8 – Lightning App

- A custom **Courier Management System Lightning App** was created.
- The app includes all important tabs such as Customers, Shipments, Agents, Branches, and Invoices.
- This allows users to access all system features in one place.
- It improves navigation and system organization.

The screenshot shows the Salesforce Lightning App Builder interface. The top navigation bar includes the URL, 'Developer Edition' label, and a 'Lightning App Builder' tab. The main workspace displays the 'App Settings' for the 'Courier Management System' app. The 'App Details & Branding' section is active, showing fields for 'App Name' (Courier Management System), 'Developer Name' (Courier_Management_System), and 'Description' (Courier Management Systems). The 'App Branding' section shows an 'Image' (a courier icon) and a 'Primary Color Hex Value' (#0070D2). The 'App Launcher Preview' section shows a preview of the app's navigation bar with the 'Courier Management System' label.

Milestone 9 – Profiles

- Profiles were created to control user permissions in the system.
- Different profiles include **Courier Admin, Branch Manager, Delivery Agent, and Customer Support.**
- Each profile has specific access rights depending on user roles.
- This ensures system security and controlled data access.



Milestone 10 – Roles

- Roles were created to define the hierarchy between users.
- The hierarchy includes **Courier Admin → Branch Manager → Delivery Agent → Customer Support.**
- Roles determine which users can view or access certain data.
- This helps maintain proper data sharing within the organization.

orgfarm-cc6b332210-dev-ed.develop.my.salesforce-setup.com/lightning/setup/Roles/home

Developer Edition

Setup Roles

Creating the Role Hierarchy

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy

Collapsible: All Expand All Show in tree view

- Global College of Arts and Science
 - Add Role
 - CEO Edit | Del | Assign
 - Add Role
 - CFO Edit | Del | Assign
 - Add Role
 - COO Edit | Del | Assign
 - Add Role
 - Courier Admin Edit | Del | Assign
 - Add Role
 - Branch Manager Edit | Del | Assign
 - Add Role
 - Customer Support Edit | Del | Assign
 - Add Role
 - Delivery Agent Edit | Del | Assign
 - Add Role
 - SVP, Customer Service & Support Edit | Del | Assign
 - Add Role

Milestone 11 – Users

- User accounts were created for different system users.
- Each user was assigned a specific **profile and role** based on their responsibilities.
- Users can log in and perform tasks according to their permissions.
- This allows multiple people to work in the system securely.

orgfarm-cc6b332210-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ManageUsers/home

Developer Edition

Setup Users

All Users

On this page you can create, view, and manage users.
To get more licenses, use the Your Account app. Let's Go

View: All Users Edit Create New View

| Action | Full Name | Alias | Username | Role | Active | Profile |
|-------------------------------|------------------|---------|--|------------------|-------------------------------------|----------------------------------|
| ☐ Edit | Chatter Expert | Chatter | chatty00d8000000024cuae.11f94vxe29f@chatter.salesforce.com | | ☒ | Chatter Free User |
| ☐ Edit | EPIC_OrgFarm | QEPIC | epic.cdp56caba35e@orgfarm.salesforce.com | | ☒ | System Administrator |
| ☐ Edit | Kumar_Ravi | kum | mmohana2006@gmail.com | Courier Admin | ☐ | Courier Admin |
| ☐ Edit | miha_madhu | mmih | rsadhana2010@gmail.com | Delivery Agent | ☒ | Delivery Agent |
| ☐ Edit | Priya_Mohana | mpdy | mmohana17@gmail.com | Customer Support | ☒ | Customer Support |
| ☐ Edit | Priya_R_Mohana | rmo | mmohana172006.a0a333c2b73d@agentforce.com | | ☒ | System Administrator |
| ☐ Edit | Reddy_Arun | arredd | arjun1.manager@cms.com | Branch Manager | ☒ | Branch Manager |
| ☐ Edit | User_Integration | integ | integration@00dd0000000024cuae.com | | ☒ | Analytics Cloud Integration User |
| ☐ Edit | User_Security | sec | insightssecurity@00dd0000000024cuae.com | | ☒ | Analytics Cloud Security User |

New User Reset Password(s) Add Multiple Users

Milestone 12 – Reports and Dashboards

- Reports and dashboards were created to analyze courier operations and business performance.
- Custom report types such as **Shipment Tracking Report, Customer Summary Report, and Agent Performance Report** were developed.
- Reports help managers view important data like shipment status, delivery performance, and revenue.
- Dashboards were designed using charts such as **donut charts, bar charts, and column charts** for better visualization.
- These dashboards provide real-time insights and help managers make better decisions.

The screenshot shows the Salesforce Setup page for Custom Report Types. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays a list of 16 custom report types, sorted by label. The table includes columns for Label, Name, Description, Category, Created By, and Created Date.

| Label | Name | Description | Category | Created By | Created Date |
|-------------------------------------|--|--|-----------------|------------|---------------------|
| Shipment Tracking Reports | Shipment_Tracking_Reports | Shipment Tracking Report | Other Report... | rmo | 3/7/2026, 1:12 A... |
| Shipment Tracking Report | Shipment_Tracking_Report | Shipment Tracking Report | Other Report... | rmo | 3/6/2026, 8:59 PM |
| Screen Flows | screen_flows_prebuilt_crt | Find out which flows get executed and how long users... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Progress Summer '24 | Program_Progress_sfdcSESv61 | Report on program progress. Specific progress on mil... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Progress Spring '24 | Program_Progress_sfdcSESv60 | Report on program progress. Specific progress on mil... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Item Progress Summer '24 | Program_Task_Progress_sfdcSESv61 | Report on tasks like exercises, milestones, and outcom... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Item Progress Spring '24 | Program_Task_Progress_sfdcSESv60 | Report on tasks like exercises, milestones, and outcom... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Definition Summer '24 | Program_Definition_sfdcSESv61 | Review your analytics with a program-like structure. Se... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Program Definition Spring '24 | Program_Definition_sfdcSESv60 | Review your analytics with a program-like structure. Se... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Orchestration Work Items Spring '24 | flow_orchestration_work_item_oottb_crt_two_four_eig... | Find out which orchestration work items were created... | Other Report... | autoproc | 3/2/2026, 7:36 PM |
| Orchestration Step Runs Spring '24 | flow_orchestration_step_run_oottb_crt_two_four_eig... | Find out which orchestration step runs were created a... | Other Report... | autoproc | 3/2/2026, 7:36 PM |

The screenshot shows the Courier Management System Dashboard. The dashboard includes three main widgets: Delivery Agents Report, Shipment Status Summary, and Revenue by Branch Report. The Delivery Agents Report shows a bar chart for Madhurai Branch with a record count of 5. The Shipment Status Summary shows a donut chart with a total record count of 5, broken down by status: Booked (2), In Transit (1), Out for Delivery (1), and Delivered (1). The Revenue by Branch Report shows a horizontal bar chart with a record count of 2 for Booked status.

| Branch | Record Count |
|----------|--------------|
| Madhurai | 5 |

| Status | Record Count |
|------------------|--------------|
| Booked | 2 |
| In Transit | 1 |
| Out for Delivery | 1 |
| Delivered | 1 |

| Status | Record Count |
|------------------|--------------|
| Booked | 2 |
| In Transit | 1 |
| Out for Delivery | 1 |
| Delivered | 1 |

Advantages

- **Centralized System:** All courier data such as customers, shipments, and invoices are stored in one platform.
- **Real-Time Tracking:** Customers and staff can track shipment status instantly.
- **Automation:** Automated flows reduce manual work and improve operational efficiency.
- **Improved Accuracy:** Validation rules help prevent incorrect or incomplete data entry.
- **Better Decision Making:** Reports and dashboards provide useful insights about deliveries and revenue.
- **Better Customer Service:** Faster updates and tracking improve customer satisfaction.

Disadvantages

- **Salesforce Dependency:** The system depends completely on the Salesforce platform.
- **Initial Setup Complexity:** Configuring objects, flows, and security settings requires technical knowledge.
- **Training Required:** Users need training to use the system effectively.
- **Internet Dependency:** The system requires an internet connection to access the cloud platform.